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A mixed-integer linear programming approach for optimal type, size and allocation of distributed generation in radial distribution systems

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ABSTRACT

This paper presents a mixed-integer linear programming approach to solving the problem of optimal type, size and allocation of distributed generators (DGs) in radial distribution systems. In the proposed formulation, (a) the steady-state operation of the radial distribution system, considering different load levels, is modeled through linear expressions; (b) different types of DGs are represented by their capability curves; (c) the short-circuit current capacity of the circuits is modeled through linear expressions; and (d) different topologies of the radial distribution system are considered. The objective function minimizes the annualized investment and operation costs. The use of a mixed-integer linear formulation guarantees convergence to optimality using existing optimization software. The results of one test system are presented in order to show the accuracy as well as the efficiency of the proposed solution technique.

1. Introduction

Distributed generation, micro-grid technologies, two-way communication systems, and demand response programs are issues that have been studied in recent years and referred to as *smart grids*. Distributed generation, when it is managed efficiently in technical and economical terms, reduces greenhouse gas emissions, improves efficiency, helps to defer system upgrades, improves reliability, and enhances energy security, among other benefits.

On the other hand, voltage increase at the end of a feeder, demand supply unbalance in a fault condition, power quality decline, increase of power losses and reduction of reliability levels are problems that may occur if the distributed generators (DGs) are not installed properly [1]. In order to solve these problems, an initial approach could be to obtain the optimal location to install the DGs by a complete enumeration of all feasible combinations of sites, types and sizes of the DGs. However, the number of alternatives could be very large as the number of variables of the problem (e.g., number of DGs, types of DGs, number of buses of the system) increases.

Therefore, to solve the problem of optimal allocation of DGs, researchers have employed conventional optimization techniques, such as generalized reduced gradient and Newton–Raphson power flows [2–5]. Analytical approaches using sensitivity factors obtained from quantities, such as the system bus impedance and admittance matrices, and the exact losses formula have also been employed [6–8]. Commercial software, which is composed of linear and non-linear solver tools, is also commonly used [9–12].

Additionally, combinatorial optimization techniques based on artificial intelligence were also used to solve the DGs allocation problem. In [13–16], evolutionary algorithms are used as the solving tool; a tabu search application is presented in [17]; and an ants colony search algorithm was performed in [18]. Although those optimization techniques can bring high quality solutions, they cannot guarantee that those solutions are globally optimal.

The major objectives considered in studies related to the allocation of DGs are to minimize system losses [2–8,14,16,17] and to minimize investment and operation costs [9–13,15,18]. However, further objectives have also been addressed, such as to minimize line loadings [2,16], to improve the voltage profile and the spinning reserve [16], and to maximize the DGs capacity with respect to the technical constraints [12]. The mathematical model of this problem is usually developed through non-linear approaches [2–11,13–15,17,18]. In few studies, for example, in [12,16], the problem is formulated linearly; however, the reactive power balance is not considered, unlike in the present work, where constraints of power balance, both real and reactive power, are considered.

The mathematical models of the DGs are usually viewed through a simple representation where the *coupling element* used to connect the DGs to the network is not detailed; that is, models of Synchronous Generators (SGs), Inverters based on Power Electronics,

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Table 1
Proposed method compared with other methods.

	Optimization model	Solution technique	Reactive power flows	Coupling elements	Load levels	Scenarios or topologies	Short-circuit currents
Proposed method	MILP	Solver AMPL	•	•	•	•	•
[6]	MINLP	Sensitivity analysis	•	○	•	○	○
[9]	MINLP	Solver GAMS	○	○	○	•	○
[10,11]	MINLP	Solver GAMS	•	○	•	○	○
[12]	MILP	Linear programming algorithm	○	○	○	○	•
[13,14]	MINLP	Evolutionary algorithm	○	○	○	○	○
[19]	MINLP	Sensitivity analysis	•	○	○	•	•
[20]	MILP	Optimization based on weight factors	○	○	○	○	•

•: Yes. ○: No.

Induction Generators (IGs), and Doubly-Fed Induction Generators (DFIGs) are disregarded [2–20]. In most cases, the type of DG (e.g., wind turbine, photovoltaic, biomass-based CHP – Combined Heat and Power – generation systems and small hydroelectric plants) is not specified [6,7,13–15].

Information regarding the DG technology is important since, according to factors such as the *coupling element* used, the operation and impact of the DGs on the system may have different results. For example, it is well known that the capability curve of an IG (commonly used in wind turbines) is different from the capability curve of a SG (generally used in biomass-based CHP generation systems and small hydroelectric plants) [21]. The ability of a DG with an SG to operate both in the reactive absorption area as well as in the reactive supply area may represent different impacts when compared with a DG with an IG, which only absorbs reactive power. DFIGs, which also allow operation in the reactive absorption and supply areas, are also commonly used; however, the capability curve and, consequently, the impact on the network are also different with regard to the IGs and the SGs [21].

Several works take into account the effect of the DGs in the short-circuit levels. In [12], sensitive indexes calculated before the optimization process were used to evaluate the contribution of increasing generation to the short-circuit currents. Another approach uses a multi-objective index that considers the impact of the DGs in the fault currents [19]. In [20], losses, voltage profile and short-circuit levels are used in the algorithm to determine the optimum sizes and locations of the DGs. Commonly, the aforementioned works only considered one topology for the system, that is, they only considered one specific configuration of the circuits that establishes the paths from the substation to the loads, despite the fact that it may change depending on the state of the system.

In Table 1, a review of the main characteristics of the proposed approach against other methods is presented.

In this paper, the problem of optimal type, size and allocation of DGs – which will be henceforth referred to only as the optimal DGs Allocation (DGA) problem – in radial distribution systems is modeled as a Mixed-Integer Linear Programming (MILP) problem. Linearizations were made to adequately represent the steady-state operation of the Electric Distribution System (EDS), considering the behavior of constant power and constant impedance type loads and different load levels. The integer nature of the decision variables represents the allocation, size and type of DGs. Different types of DGs are represented by their capability curves.

The objective of the optimization problem is to minimize the total investment and operation costs subject to operational and physical constraints. The short-circuit current capacity of the circuits is considered and, to account for changes in demand, considerations such as annual variation of the demand (represented through load levels) and different possible topologies for the distribution system are included in the model. In addition, such modeling allows incorporation of probabilistic scenarios for the loads, each with an occurrence probability. The proposed model was tested in a system of 33 buses.

The optimization process is developed from the standpoint of the utility. From this point of view, there are two possibilities:

1. DGs are owned by the utility, in which case the utility is free to allocate them in places it deems advisable according to the solutions of the optimization process.
2. DGs are not owned by the utility, in which case the utility can provide incentives to install DGs in locations that are appropriate for the system, also in accordance with the solutions of an optimization process.

Although DGs are mainly managed by private companies, there are also real applications in which DGs are owned by the utility. As presented in [22], different distributed generation technologies are implemented to fulfill the requirements of a wide range of applications according to the load requirements. In a base load application, DGs owned by the utility are commonly used to provide part of the main required power and support the system by enhancing the voltage profile, reducing the power losses and improving the power quality. According to this definition, our proposal is framed within a base load application, in which DGs are owned by the utility. Therefore, the utility is free to allocate DGs in places it deems advisable for the system.

The proposed method is a useful tool for distribution systems planning, which presents great advantages, in enabling the planner to take appropriate decisions in accordance with the characteristics of the system under study and the economic resources available. It allows the planner to carry out a suitable planning of the distribution system aimed at situations where there are economic opportunities for the installation of DGs for generating power near load centers. The main contributions of this paper are as follows:

1. A novel model for the steady-state operation of a radial EDS through the use of linear expressions.
2. An MILP formulation for the optimal DGA in radial EDSs that presents an efficient computational behavior with conventional MILP solvers. In this formulation, *coupling elements* widely used in distributed generation are considered. In addition, constraints necessary to ensure that DGs allocated in the system will not cause an increase in the short-circuit currents beyond the capacity of the system are also included.

2. The distributed generators allocation problem in radial distribution systems

2.1. Assumptions

In order to represent the steady-state operation of a radial EDS, the following assumptions are made:

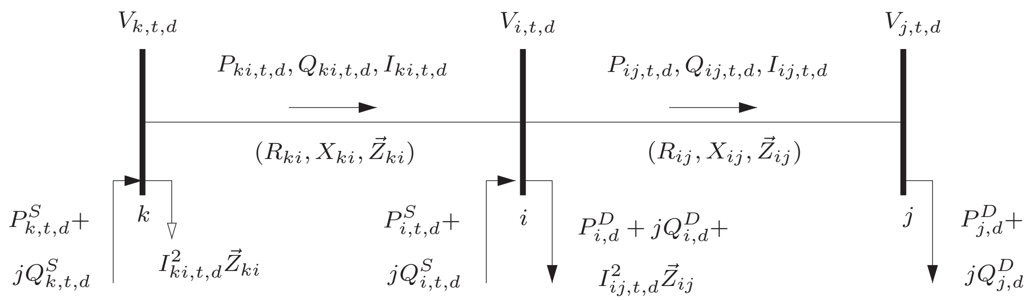

Fig. 1. Illustrative example.

1. The load is represented as constant real and reactive power, and constant impedance.
2. In branch ij node i is closer to the substation node than node j.
3. The real and reactive power losses on branch ij are concentrated in node i.
4. The EDS is balanced and represented by a monophasic equivalent.

The four considerations are shown in Fig. 1, where $R_{ij}I^2_{ij,t,d}$ and $X_{ij}I^2_{ij,t,d}$ are the real and reactive power losses of branch ij, respectively, represented by $I^2_{ij,t,d}\vec{Z}_{ij}$. In this figure, R_{ij}, X_{ij} and Z_{ij} are, respectively, resistance, reactance and impedance of branch ij; $P_{ij,t,d}$, $Q_{ij,t,d}$ and $I_{ij,t,d}$, are, respectively, real power flow, reactive power flow and current flow magnitude of branch ij of topology t in load level d; $P^S_{i,t,d}$ and $Q^S_{i,t,d}$ are, respectively, real and reactive power provided by substation at node i of topology t in load level d; $P^D_{i,d}$ and $Q^D_{i,d}$ are, respectively, real and reactive power demanded at node i in load level d; and $V_{i,t,d}$ is the voltage magnitude at node i of topology t in load level d.

2.2. Steady-state operation of a radial distribution system

The equations that represent the steady-state operation of a radial distribution system are shown in (1) and (2). These equations are frequently used in the load flow sweep method [23].

$$\sum_{ki\in\Omega_l} P_{ki,t,d} - \sum_{ij\in\Omega_l}(P_{ij,t,d} + R_{ij}I^2_{ij,t,d}) + P^S_{i,t,d} = P^D_{i,d} \quad \forall i \in \Omega_b, \forall t \in \Omega_t, \forall d \in \Omega_d \tag{1a}$$

$$\sum_{ki\in\Omega_l} Q_{ki,t,d} - \sum_{ij\in\Omega_l}(Q_{ij,t,d} + X_{ij}I^2_{ij,t,d}) + Q^S_{i,t,d} = Q^D_{i,d} \quad \forall i \in \Omega_b, \forall t \in \Omega_t, \forall d \in \Omega_d \tag{1b}$$

$$V^2_{i,t,d} - 2(R_{ij}P_{ij,t,d} + X_{ij}Q_{ij,t,d}) - Z^2_{ij}I^2_{ij,t,d} - V^2_{j,t,d} = 0 \times \quad \forall ij \in \Omega_l, \forall t \in \Omega_t, \forall d \in \Omega_d \tag{2a}$$

$$I^2_{ij,t,d} = \frac{P^2_{ij,t,d} + Q^2_{ij,t,d}}{V^2_{j,t,d}} \quad \forall ij \in \Omega_l, \forall t \in \Omega_t, \forall d \in \Omega_d \tag{2b}$$

where Ω_l, Ω_b, Ω_t and Ω_d are, respectively, sets of branches, nodes, topologies and load levels.

Eqs. (1a) and (1b) are the conventional representation of the load balance (see Fig. 1). Through (2a), it is possible to obtain the voltage magnitude of the final node ($V_{j,t,d}$) in terms of the voltage magnitude of the initial node ($V_{i,t,d}$), the real power flow ($P_{ij,t,d}$), the reactive power flow ($Q_{ij,t,d}$), the current magnitude ($I_{ij,t,d}$) and the electrical parameters of branch ij. Finally, the square current magnitude is calculated using (2b). The above equations are also valid in the presence of distributed generation.

2.3. Modeling the distributed generator power limits

In this work, the *coupling elements* used to connect the DGs to the network are represented by their capability curves, which limit the operation of the DGs in function of the real and reactive power generated. The capability curves of SGs (see Fig. 2(a)) and DFIGs (see Fig. 2(b)) are considered in this work because they are widely used in distributed generation, mainly in wind turbines, biomass-based CHP generation systems and small hydroelectric plants. The capability curve of SG presented in Fig. 2(a) is based on that presented in [24] and the capability curve of DFIG presented in Fig. 2(b) is based on that presented in [25].

Based on the generation limits depicted in Fig. 2(a) and (b), the points ($Q_{1,g}$,$P_{1,g}$), ($Q_{2,g}$,$P_{2,g}$), ($Q_{3,g}$,$P_{3,g}$) and ($Q_{4,g}$,$P_{4,g}$) are defined. For the SG, ($Q_{1,g}$,$P_{1,g}$) is the intersection between the under-excitation and armature current limits, ($Q_{2,g}$,$P_{2,g}$) is the intersection between the armature current limit and the P axis, ($Q_{3,g}$,$P_{3,g}$) is the half of the arc of the armature current limit between points ($Q_{2,g}$,$P_{2,g}$) and ($Q_{4,g}$,$P_{4,g}$), and ($Q_{4,g}$,$P_{4,g}$) is the intersection between the armature current and field current limits. For the DFIG, ($Q_{1,g}$,$P_{1,g}$) is the half of the arc of the armature current limit between points ($\underline{Q}_g$,0) and ($Q_{2,g}$,$P_{2,g}$), ($Q_{2,g}$,$P_{2,g}$) is the intersection between the armature current and field current limits, ($Q_{3,g}$,$P_{3,g}$) is the intersection between the field current limit and the P axis, and ($Q_{4,g}$,$P_{4,g}$) is the half of the arc of the field current limit between points ($Q_{3,g}$,$P_{3,g}$) and ($\overline{Q}_g$,0). Because the capability curves of the SGs and DFIGs are non-linear functions, and with the aim of obtaining linear expressions that represent the operation limits of the DGs, the capability curves can be linearized using the expressions shown in (3) based on the aforementioned points.

$$P^{DG}_{i,t,d,g} \le \frac{P_{1,g}}{Q_{1,g} - \underline{Q}_g}(Q^{DG}_{i,t,d,g} - \underline{Q}_g) \quad \forall i \in \Omega_b, \forall t \in \Omega_t, \forall d \in \Omega_d, \forall g \in \Omega_g \tag{3a}$$

$$P^{DG}_{i,t,d,g} \le \frac{P_{2,g} - P_{1,g}}{Q_{2,g} - Q_{1,g}}(Q^{DG}_{i,t,d,g} - Q_{2,g}) + P_{2,g} \quad \forall i \in \Omega_b, \forall t \in \Omega_t, \forall d \in \Omega_d, \forall g \in \Omega_g \tag{3b}$$

$$P^{DG}_{i,t,d,g} \le \frac{P_{3,g} - P_{2,g}}{Q_{3,g} - Q_{2,g}}(Q^{DG}_{i,t,d,g} - Q_{3,g})$$

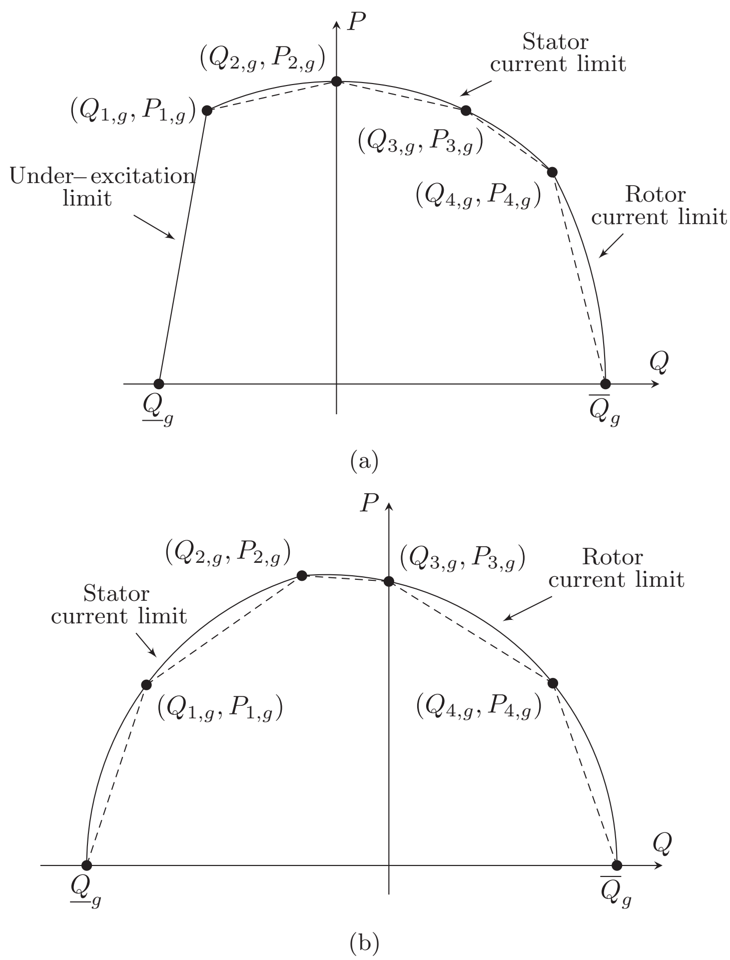

Fig. 2. Capability curves: (a) synchronous generator and (b) doubly fed induction generator.

$$+P_{3,g} \quad \forall i \in \Omega_b, \forall t \in \Omega_t, \forall d \in \Omega_d, \forall g \in \Omega_g \tag{3c}$$

$$P^{DG}_{i,t,d,g} \leq \frac{P_{4,g} - P_{3,g}}{Q_{4,g} - Q_{3,g}}(Q^{DG}_{i,t,d,g} - Q_{4,g}) + P_{4,g} \quad \forall i \in \Omega_b, \forall t \in \Omega_t, \forall d \in \Omega_d, \forall g \in \Omega_g \tag{3d}$$

$$P^{DG}_{i,t,d,g} \leq \frac{P_{4,g}}{Q_{4,g} - \overline{Q}_g}(Q^{DG}_{i,t,d,g} - \overline{Q}_g) \times \forall i \in \Omega_b, \forall t \in \Omega_t, \forall d \in \Omega_d, \forall g \in \Omega_g \tag{3e}$$

where Ω_g is the set of DG types; $P^{DG}_{i,t,d,g}$ and $Q^{DG}_{i,t,d,g}$ are, respectively, real and reactive power provided by DG of type g at node i of topology t in load level d; $P_{k,g}$ and $Q_{k,g}$ represent the kth point of the linearization of the capability curve of DG of type g; and $\overline{Q}_g$ and $\underline{Q}_g$ are, respectively, maximum and minimum reactive power limits of DG of type g.

2.4. Short-circuit analysis considering distributed generation

Allocating DGs in the system affects the voltage profile, the branch currents, as well as the short-circuit currents. It must be verified that the DGs allocated in the system will not cause an increase of the short-circuit currents beyond the short-circuit current capacity of the circuits. So, in this section, the necessary constraints to ensure compliance of the short-circuit current capacity of the circuits are presented.

For short-circuit analysis, the following assumptions are made:

1. The substation is modeled as a power source with a voltage of $1\angle 0^{\circ} pu$ connected through an equivalent impedance of the upstream network.
2. The DG is modeled as a power source with voltage of $1\angle 0^{\circ} pu$ behind a subtransitory reactance.
3. The faults are symmetrical three-phase short-circuits, and only one occurs at a time.
4. Considering that the currents in short-circuit are much larger than the load currents in steady-state, the loads are ignored, that is, the load currents are neglected.

The equivalent impedance of the system can be calculated using the short-circuit capacity at the substation. If there is a fault at bus f for the system topology t, the fault voltages are calculated by solving the equation system (4),

$$(Y^{BUS}_t + Y^{DG})\widetilde{V}^{BUS}_{f,t} = \widetilde{I}^{BUS}_{f,t} \quad \forall f \in \Omega_b, \forall t \in \Omega_t \tag{4}$$

where Y^{BUS}_t is the nodal admittance matrix that has the information of the branches admittance and the external equivalent admittance; Y^{DG} is the diagonal matrix conformed by the admittances of the DGs; $\widetilde{V}^{BUS}_{f,t}$ is the vector of fault voltages; and $\widetilde{I}^{BUS}_{f,t}$ is the vector of current injections.

If the position of the DGs depends on the binary variable $y_{i,g}$ ($i \in \Omega_b$ and $g \in \Omega_g$), the second term of the left member of (4) is composed by the product of this binary variable, the admittances of the DGs and the fault voltages. The ith element of that vector is

$\sum_{g\in\Omega_g}(y_{i,g}/jX_g^{DG})\widetilde{V}_{f,i,t}$, which is separated into its real and imaginary parts in (5). Note that the terms in (5) have the product of two variables. So, aiming to obtain a set of linear equations for the short-circuit analysis, those terms are linearized in (6) using $W^{re}_{f,i,t,g}$ and $W^{im}_{f,i,t,g}$. Constraints (6) are a way to represent the product of the binary variable $y_{i,g}$ with the voltage $\widetilde{V}^{im}_{f,i,t}$, using disjunctive constraints. Therefore, if $y_{i,g}$ is zero, then, by (6a), $W^{re}_{f,i,t,g}$ is zero; and, if $y_{i,g}$ is one, then, by (6b), $W^{re}_{f,i,t,g}$ is $-\widetilde{V}^{im}_{f,i,t}/X_g^{DG}$.

$$\sum_{g\in\Omega_g}\frac{y_{i,g}}{jX_g^{DG}}\widetilde{V}_{f,i,t} = -\sum_{g\in\Omega_g}\frac{y_{i,g}}{X_g^{DG}}\widetilde{V}^{im}_{f,i,t} + j\sum_{g\in\Omega_g}\frac{y_{i,g}}{X_g^{DG}}\widetilde{V}^{re}_{f,i,t} \quad \forall f,i\in\Omega_b,\forall t\in\Omega_t \tag{5}$$

$$\left|W^{re}_{f,i,t,g}\right| \le \frac{\overline{V}}{X_g^{DG}}y_{i,g} \quad \forall f,i\in\Omega_b,\forall t\in\Omega_t,\forall g\in\Omega_g \tag{6a}$$

$$\left|-\frac{\widetilde{V}^{im}_{f,i,t}}{X_g^{DG}} - W^{re}_{f,i,t,g}\right| \le \frac{\overline{V}}{X_g^{DG}}(1-y_{i,g}) \quad \forall f,i\in\Omega_b,\forall t\in\Omega_t,\forall g\in\Omega_g \tag{6b}$$

$$\left|W^{im}_{f,i,t,g}\right| \le \frac{\overline{V}}{X_g^{DG}}y_{i,g} \quad \forall f,i\in\Omega_b,\forall t\in\Omega_t,\forall g\in\Omega_g \tag{6c}$$

$$\left|\frac{\widetilde{V}^{re}_{f,i,t}}{X_g^{DG}} - W^{im}_{f,i,t,g}\right| \le \frac{\overline{V}}{X_g^{DG}}(1-y_{i,g}) \quad \forall f,i\in\Omega_b,\forall t\in\Omega_t,\forall g\in\Omega_g \tag{6d}$$

where $y_{i,g}$ is the binary variable for allocation of a DG of type g at node i; X_g^{DG} is the reactance of DG of type g; $\widetilde{V}_{f,i,t}$ is the voltage at node i of topology t for a fault at bus f; $\widetilde{V}^{re}_{f,i,t}$ and $\widetilde{V}^{im}_{f,i,t}$ are, respectively, real and imaginary parts of $\widetilde{V}_{f,i,t}$; $W^{re}_{f,i,t,g}$ and $W^{im}_{f,i,t,g}$ are, respectively, variables used to linearize $y_{i,g}\widetilde{V}^{re}_{f,i,t}/X_g^{DG}$ and $y_{i,g}\widetilde{V}^{im}_{f,i,t}/X_g^{DG}$; and $\overline{V}$ is the maximum voltage magnitude.

Using the variables $W^{re}_{f,i,t,g}$ and $W^{im}_{f,i,t,g}$, a linear version of the real and imaginary parts of (4) is presented in (7) and (8). The variables $\widetilde{I}^{re}_{f,i,t}$ and $\widetilde{I}^{im}_{f,i,t}$ (real and imaginary parts of the injected current at node i of topology t for a fault at bus f, respectively) are equal to zero $\forall f \neq i$.

$$\sum_{ij\in\Omega_l}\left[\frac{R_{ij}}{Z_{ij}^2}\left(\widetilde{V}^{re}_{f,i,t}-\widetilde{V}^{re}_{f,j,t}\right)+\frac{X_{ij}}{Z_{ij}^2}\left(\widetilde{V}^{im}_{f,i,t}-\widetilde{V}^{im}_{f,j,t}\right)\right]+\sum_{ji\in\Omega_l}\left[\frac{R_{ji}}{Z_{ji}^2}\left(\widetilde{V}^{re}_{f,i,t}-\widetilde{V}^{re}_{f,j,t}\right)+\frac{X_{ji}}{Z_{ji}^2}\left(\widetilde{V}^{im}_{f,i,t}-\widetilde{V}^{im}_{f,j,t}\right)\right] + \sum_{g\in\Omega_g}W^{re}_{f,i,t,g} = \widetilde{I}^{re}_{f,i,t} \quad \forall f,i\in\Omega_b,\forall t\in\Omega_t \tag{7}$$

$$\sum_{ij\in\Omega_l}\left[\frac{R_{ij}}{Z_{ij}^2}\left(\widetilde{V}^{im}_{f,i,t}-\widetilde{V}^{im}_{f,j,t}\right)-\frac{X_{ij}}{Z_{ij}^2}\left(\widetilde{V}^{re}_{f,i,t}-\widetilde{V}^{re}_{f,j,t}\right)\right]+\sum_{ji\in\Omega_l}\left[\frac{R_{ji}}{Z_{ji}^2}\left(\widetilde{V}^{im}_{f,i,t}-\widetilde{V}^{im}_{f,j,t}\right)-\frac{X_{ji}}{Z_{ji}^2}\left(\widetilde{V}^{re}_{f,i,t}-\widetilde{V}^{re}_{f,j,t}\right)\right] + \sum_{g\in\Omega_g}W^{im}_{f,i,t,g} = \widetilde{I}^{im}_{f,i,t} \quad \forall f,i\in\Omega_b,\forall t\in\Omega_t \tag{8}$$

For a fault at bus f, the short-circuit capacity of the circuits must be satisfied. It is assumed that the capacity of the conductors is monotonously decreasing from the substation to the location of the fault, and that the substation provides the bigger part in the short-circuit current. Then, the circuit that holds the bigger fault current is the one that connects the bus f with the upstream network (circuit kf). So, for a fault at bus f, the short-circuit current at circuit kf must be less than its short-circuit capacity. The real and imaginary parts of the short-circuit current at circuit kf ($Icc^{re}_{f,kf,t}$ and $Icc^{im}_{f,kf,t}$, respectively) are calculated by using (9).

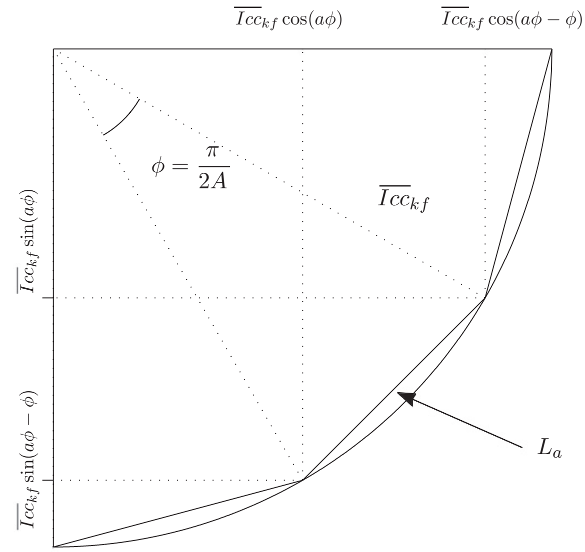

Fig. 3. Constraints for short-circuit currents.

$$Icc^{re}_{f,kf,t} = \frac{R_{kf}}{Z_{kf}^2}\left(\widetilde{V}^{re}_{f,k,t}-\widetilde{V}^{re}_{f,f,t}\right)+\frac{X_{kf}}{Z_{kf}^2}\left(\widetilde{V}^{im}_{f,k,t}-\widetilde{V}^{im}_{f,f,t}\right) \times kf\in\Omega_l,\forall f\in\Omega_b,\forall t\in\Omega_t \tag{9a}$$

$$Icc^{im}_{f,kf,t} = -\frac{X_{kf}}{Z_{kf}^2}\left(\widetilde{V}^{re}_{f,k,t}-\widetilde{V}^{re}_{f,f,t}\right)+\frac{R_{kf}}{Z_{kf}^2}\left(\widetilde{V}^{im}_{f,k,t}-\widetilde{V}^{im}_{f,f,t}\right) \times kf\in\Omega_l,\forall f\in\Omega_b,\forall t\in\Omega_t \tag{9b}$$

Because the short-circuit analysis involves power sources with a voltage of $1\angle 0^\circ pu$ and impedance elements with a positive angle, the short-circuit currents are lagging behind the voltage and therefore $Icc^{re}_{f,kf,t} \ge 0$ and $Icc^{im}_{f,kf,t} \le 0$. Then, the real and imaginary parts of the short-circuit current must be limited in a circle section of radius $\overline{Icc}_{kf}$ in the forth quadrant. For this purpose, a set of A constraints, each one associated with a line segment L_a, is established in (10) (see Fig. 3).

$$Icc^{im}_{f,kf,t} \ge m_a^{Icc}(Icc^{re}_{f,kf,t} - \overline{Icc}_{kf}\cos(a\phi)) - \overline{Icc}_{kf}\sin(a\phi) \quad a=1,\ \ldots,\ A,\ kf\in\Omega_l,\forall f\in\Omega_b,\forall t\in\Omega_t \tag{10}$$

where $m_a^{Icc} = (-\sin(a\phi) + \sin(a\phi - \phi))/(\cos(a\phi) - \cos(a\phi - \phi))$ and $-Icc_{f,kf,t}^{re}, Icc_{f,kf,t}^{im} \leq 0$

In Eq. (10), m_a^{Icc} is the slope of the ath line segment for the short-circuit constraints; $\overline{Icc}_{kf}$ is the maximum short-circuit current magnitude of branch kf; and A is the number of line segments for linearization of the short-circuit constraints.

If the capacity of the conductors is not monotonously decreasing from the substation to the location of the fault as it was assumed, then, instead of branch kf, it is chosen the branch with the lower short-circuit current capacity between those in the path from the substation to the location of the fault.

2.5. Non-linear formulation for the distributed generators allocation problem

The DGA problem can be modeled as a Mixed-Integer Non-Linear Programming (MINLP) problem as follows:

$$min \sum_{i\in\Omega_b}\sum_{g\in\Omega_g} c_g y_{i,g} + \sum_{i\in\Omega_b}\sum_{t\in\Omega_t}\sum_{d\in\Omega_d}\alpha_{t,d} ec_d^S P_{i,t,d}^S + \sum_{i\in\Omega_b}\sum_{t\in\Omega_t}\sum_{d\in\Omega_d}\sum_{g\in\Omega_g}\alpha_{t,d} ec_d^g P_{i,t,d,g}^{DG} \quad (11)$$

subject to
constraints (3), (6)–(10),

$$\sum_{ki\in\Omega_l} P_{ki,t,d} - \sum_{ij\in\Omega_l}(P_{ij,t,d} + R_{ij} I_{ij,t,d}^{sqr}) + P_{i,t,d}^S + \sum_{g\in\Omega_g} P_{i,t,d,g}^{DG} = P_{i,d}^D + P_{i,d}^{Dz}\left(\frac{V_{i,t,d}}{V_b}\right)^2 \quad \forall i \in \Omega_b, \forall t \in \Omega_t, \forall d \in \Omega_d \quad (12)$$

$$\sum_{ki\in\Omega_l} Q_{ki,t,d} - \sum_{ij\in\Omega_l}(Q_{ij,t,d} + X_{ij} I_{ij,t,d}^{sqr}) + Q_{i,t,d}^S + \sum_{g\in\Omega_g} Q_{i,t,d,g}^{DG} = Q_{i,d}^D + Q_{i,d}^{Dz}\left(\frac{V_{i,t,d}}{V_b}\right)^2 \quad \forall i \in \Omega_b, \forall t \in \Omega_t, \forall d \in \Omega_d \quad (13)$$

$$V_{i,t,d}^{sqr} - 2(R_{ij}P_{ij,t,d} + X_{ij}Q_{ij,t,d}) - Z_{ij}^2 I_{ij,t,d}^{sqr} - V_{j,t,d}^{sqr} = 0 \times \forall ij \in \Omega_l, \forall t \in \Omega_t, \forall d \in \Omega_d \quad (14)$$

$$V_{j,t,d}^{sqr} I_{ij,t,d}^{sqr} = P_{ij,t,d}^2 + Q_{ij,t,d}^2 \quad \forall ij \in \Omega_l, \forall t \in \Omega_t, \forall d \in \Omega_d \quad (15)$$

$$\underline{V}^2 \leq V_{i,t,d}^{sqr} \leq \overline{V}^2 \quad \forall i \in \Omega_b, \forall t \in \Omega_t, \forall d \in \Omega_d \quad (16)$$

$$0 \leq I_{ij,d}^{sqr} \leq \overline{I}_{ij,t}^2 \quad \forall ij \in \Omega_l, \forall t \in \Omega_t, \forall d \in \Omega_d \quad (17)$$

$$0 \leq P_{i,t,d,g}^{DG} \leq \overline{P_g} y_{i,g} \quad \forall i \in \Omega_b, \forall t \in \Omega_t, \forall d \in \Omega_d, \forall g \in \Omega_g \quad (18)$$

$$\underline{Q_g} y_{i,g} \leq Q_{i,t,d,g}^{DG} \leq \overline{Q_g} y_{i,g} \quad \forall i \in \Omega_b, \forall t \in \Omega_t, \forall d \in \Omega_d, \forall g \in \Omega_g \quad (19)$$

$$P_{i,t,d}^S tan(\arccos(\underline{pf}^S)) \leq \left|Q_{i,t,d}^S\right| \quad \forall i \in \Omega_b, \forall t \in \Omega_t, \forall d \in \Omega_d \quad (20)$$

$$\sum_{i\in\Omega_b}\sum_{g\in\Omega_g} y_{i,g} \leq \overline{n}_{dg} \quad (21)$$

$$\sum_{g\in\Omega_g} y_{i,g} \leq 1 \quad \forall i \in \Omega_b \quad (22)$$

$$y_{i,g} \in \{0, 1\} \quad \forall i \in \Omega_b, \forall g \in \Omega_g \quad (23)$$

where c_g is the annualized installation cost of DG of type g (in US\$); $\alpha_{t,d}$ is the number of hours in one year of topology t in load level d; ec_d^S and ec_d^g are, respectively, energy cost for the substation and for DG of type g (in US\$/kWh) in load level d; $I_{ij,t,d}^{sqr}$ is the square of $I_{ij,t,d}$; $P_{i,d}^{Dz}$ and $Q_{i,d}^{Dz}$ are, respectively, real and reactive power demanded by a constant impedance load at node i in load level d; V_b is the magnitude of the base voltage; $V_{i,t,d}^{sqr}$ is the square of $V_{i,t,d}$; $\underline{V}$ is the minimum voltage magnitude; $\overline{I}_{ij}$ is the maximum current magnitude of branch ij; $\overline{P}_g$ is the maximum real power limit of DG of type g; $\underline{pf}^S$ is the minimum leading and lagging substation power factor; and $\overline{n}_{dg}$ is the maximum number of DGs that can be installed in the system.

In the non-linear formulation presented above, the variables $I_{ij,d}^{sqr}$ and $V_{i,d}^{sqr}$ are used to represent $I_{ij,d}^2$ and $V_{i,d}^2$, respectively. The objective function (11) is the annualized investment and operation costs. The first part represents the investment cost (installation of DGs), the second part represents the substation annual operation cost and the third part is the DGs annual operation cost. The set of Eqs. (3) represents the capability curves of the DGs. Eqs. (6)–(10) are the linear representation of the short-circuit constraints. Eqs. (12)–(15) represent the steady-state operation, and they are a natural extension of (1) and (2), considering the presence of DGs. Eqs. (12) and (13) take into account constant power and constant impedance type loads. Eq. (16) represents the constraints of the square voltage magnitude of the nodes, while (17) represents the limit of the flows of current in branch ij. Eqs. (18) and (19) limit the power of the DGs according to the associated investment binary variable. The power factor of the substation, both leading and lagging, must be within specified limits, as shown in (20). The maximum number of DGs that can be installed in the system is represented by (21). Eq. (22) assures that duplication of DGs is not allowed. Eq. (23) represents the binary nature of the installation of DGs. A DG is installed if the corresponding value of $y_{i,g}$ is equal to one and is not installed if it is equal to zero. The binary investment variable $y_{i,g}$ is the decision variable, and a feasible operation solution for the distribution system depends on its value. The remaining variables represent the operating state of a feasible solution. For a feasible investment proposal, defined through specified values of $y_{i,g}$, several feasible operation states are possible, and the optimization technique finds the optimal solution.

Note that (11)–(14) and (16)–(22) are linear expressions, but (15) contains square terms and the product of two variables. With the aim of using a conventional MILP solver, it is desirable to obtain a linear equivalent for constraint (15).

2.6. Linearization

The left member of (15) is linearized by discretization of $V_{j,t,d}^{sqr}$ using the binary variables $x_{j,t,d,s} \forall s = 1 \ldots S$, where $x_{j,t,d,s} = 1$ if $V_{j,t,d}^{sqr}$ is greater than $\underline{V}^2 + s\overline{\Delta}^V$, as presented in Fig. 4. This condition is modeled in (24), which shows how the variables $x_{j,t,d,s}$ are calculated.

$$\underline{V}^2 + \sum_{s=1}^{S}\left(x_{j,t,d,s}\overline{\Delta}^V\right) \leq V_{j,t,d}^{sqr} \leq \underline{V}^2 + \overline{\Delta}^V + \sum_{s=1}^{S}\left(x_{j,t,d,s}\overline{\Delta}^V\right) \quad \forall j \in \Omega_b, \forall t \in \Omega_t, \forall d \in \Omega_d \quad (24a)$$

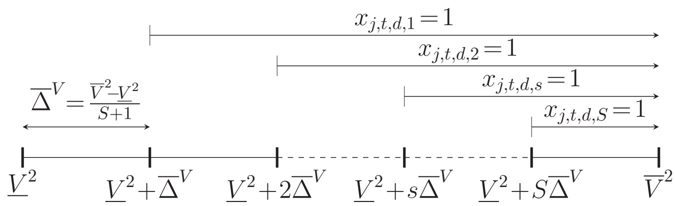

Fig. 4. Discretization of the square voltage magnitude.

$$x_{j,t,d,s} \le x_{j,t,d,s-1} \quad \forall s = 2 \ldots S, \forall j \in \Omega_b, \forall t \in \Omega_t, \forall d \in \Omega_d \tag{24b}$$

$$x_{j,t,d,s} \in \{0, 1\} \quad \forall s = 2 \ldots S, \forall j \in \Omega_b, \forall t \in \Omega_t, \forall d \in \Omega_d \tag{24c}$$

where S is the number of discretizations and $\overline{\Delta}^V$ is the discretization step of the variable $V^{sqr}_{i,t,d}$.

The product $V^{sqr}_{j,t,d} I^{sqr}_{ij,t,d}$ is calculated using the middle point of the first interval of the discretization of the square magnitude voltage multiplied by the square current flow magnitude plus the successive power corrections ($P^c_{j,t,d,s}$) that depend on $\overline{\Delta}^V$, $I^{sqr}_{ij,t,d}$ and $x_{j,t,d,s}$, according to (25) and (26).

$$V^{sqr}_{j,t,d} I^{sqr}_{ij,t,d} = \left(\underline{V}^2 + \frac{1}{2}\overline{\Delta}^V\right) I^{sqr}_{ij,t,d} + \sum_{s=1}^{S} P^c_{j,t,d,s}$$
$$\times \forall ij \in \Omega_l, \forall t \in \Omega_t, \forall d \in \Omega_d \tag{25}$$

$$0 \le \overline{\Delta}^V I^{sqr}_{ij,t,d} - P^c_{j,t,d,s} \le \overline{\Delta}^V \bar{I}_{ij}^{\,2} (1 - x_{j,t,d,s})$$
$$\times \forall ij \in \Omega_l, \forall t \in \Omega_t, \forall d \in \Omega_d, s = 1 \ldots S \tag{26a}$$

$$0 \le P^c_{j,t,d,s} \le \overline{\Delta}^V \bar{I}_{ij}^{\,2} x_{j,t,d,s} \quad \forall ij \in \Omega_l, \forall t \in \Omega_t, \forall d \in \Omega_d, s = 1 \ldots S \tag{26b}$$

where $P^c_{j,t,d,s}$ are power corrections used in the discretization of $V^{sqr}_{j,t,d} I^{sqr}_{ij,t,d}$.

Constraint (25) is a linear approximation of the product of $V^{sqr}_{j,t,d}$ and $I^{sqr}_{ij,t,d}$. Constraint (26) define the values of $P^c_{j,t,d,s}$. If $x_{j,t,d,s} = 0$, then $P^c_{j,t,d,s} = 0$ and $I^{sqr}_{ij,t,d} \le \bar{I}^2_{ij}$; otherwise, $P^c_{j,t,d,s} = \overline{\Delta}^V I^{sqr}_{ij,t,d}$ and $P^c_{j,t,d,s} \le \overline{\Delta}^V \bar{I}^2_{ij}$, where $\overline{\Delta}^V \bar{I}^2_{ij}$ plays exactly the role of the "Big M" factor and provides a sufficient degree of freedom to $P^c_{j,t,d,s}$. The right member of (15) is linearized as described in [26] and defined in (27) and (28).

$$P^2_{ij,t,d} + Q^2_{ij,t,d} = \sum_{r=1}^{R} m^S_{ij,r} \Delta^P_{ij,t,d,r} + \sum_{r=1}^{R} m^S_{ij,t,r} \Delta^Q_{ij,t,d,r}$$
$$\times \forall ij \in \Omega_l, \forall t \in \Omega_t, \forall d \in \Omega_d \tag{27}$$

$$P^+_{ij,t,d} - P^-_{ij,t,d} = P_{ij,t,d} \quad \forall ij \in \Omega_l, \forall t \in \Omega_t, \forall d \in \Omega_d \tag{28a}$$

$$Q^+_{ij,t,d} - Q^-_{ij,t,d} = Q_{ij,t,d} \quad \forall ij \in \Omega_l, \forall t \in \Omega_t, \forall d \in \Omega_d \tag{28b}$$

$$P^+_{ij,t,d} + P^-_{ij,t,d} = \sum_{r=1}^{R} \Delta^P_{ij,t,d,r} \quad \forall ij \in \Omega_l, \forall t \in \Omega_t, \forall d \in \Omega_d \tag{28c}$$

$$Q^+_{ij,t,d} + Q^-_{ij,t,d} = \sum_{r=1}^{R} \Delta^Q_{ij,t,d,r} \quad \forall ij \in \Omega_l, \forall t \in \Omega_t, \forall d \in \Omega_d \tag{28d}$$

$$0 \le \Delta^P_{ij,t,d,r} \le \overline{\Delta}^S_{ij} \quad \forall r = 1 \ldots R, \forall ij \in \Omega_l, \forall t \in \Omega_t, \forall d \in \Omega_d \tag{28e}$$

$$0 \le \Delta^Q_{ij,t,d,r} \le \overline{\Delta}^S_{ij} \quad \forall r = 1 \ldots R, \forall ij \in \Omega_l, \forall t \in \Omega_t, \forall d \in \Omega_d \tag{28f}$$

$$0 \le P^+_{ij,t,d}, P^-_{ij,t,d}, Q^+_{ij,t,d}, Q^-_{ij,t,d} \quad \forall ij \in \Omega_l, \forall t \in \Omega_t, \forall d \in \Omega_d \tag{28g}$$

where R is the number of blocks of the piecewise linearization; $m^S_{ij,r}$ is the slope of the rth block of the power flow of branch ij; $\Delta^P_{ij,t,d,r}$ and $\Delta^Q_{ij,t,d,r}$ are, respectively, values of the rth block of $|P_{ij,t,d}|$ and $|Q_{ij,t,d}|$; and $\overline{\Delta}^S_{ij}$ is the upper bound of each block of the power flow of branch ij. In the above equations, $m^S_{ij,r} = (2r-1)\overline{\Delta}^S_{ij}$ and $\overline{\Delta}^S_{ij} = \overline{V}\bar{I}_{ij}/R$, $\forall ij \in \Omega_l$ and $r = 1 \ldots R$.

Eqs. (27) and (28) are a set of linear expressions of the right member of (15) and $m^S_{ij,r}$ and $\overline{\Delta}^S_{ij}$ are constant parameters. $\sum_{r=1}^{R} m^S_{ij,r} \Delta^P_{ij,t,d,r}$ and $\sum_{r=1}^{R} m^S_{ij,r} \Delta^Q_{ij,t,d,r}$ are the linear approximations of P^2_{ij} and Q^2_{ij}, respectively. The linearization of P^2_{ij} is shown in Fig. 5. Variables $P^+_{ij,t,d}$ and $P^-_{ij,t,d}$ are non-negative auxiliaries to obtain $|P_{ij}|$, as shown in (28a). Variables $Q^+_{ij,t,d}$ and $Q^-_{ij,t,d}$ are non-negative auxiliaries to obtain $|Q_{ij}|$, as shown in (28b). Constraints (28c) and (28d) state that $|P_{ij,t,d}|$ and $|Q_{ij,t,d}|$ are equal to the sum of the values in each block of the discretization, respectively. Constraints (28e) and (28f) set the upper and lower limits of the contribution of each block of $|P_{ij,t,d}|$ and $|Q_{ij,t,d}|$, respectively.

2.7. MILP formulation for the distributed generators allocation problem

The DGA problem could be modeled as a MILP problem as follows:

$$\min \sum_{i\in\Omega_b}\sum_{g\in\Omega_g} c_g y_{i,g} + \sum_{i\in\Omega_b}\sum_{t\in\Omega_t}\sum_{d\in\Omega_d} \alpha_{t,d} ec^S_d P^S_{i,t,d}$$
$$+ \sum_{i\in\Omega_b}\sum_{t\in\Omega_t}\sum_{d\in\Omega_d}\sum_{g\in\Omega_g} \alpha_{t,d} ec^g_d P^{DG}_{i,t,d,g} \tag{29}$$

subject to
constraints (3), (6)–(10), (12)–(14), (16)–(24), (26), (28),

$$(\underline{V}^2 + \frac{1}{2}\overline{\Delta}^V) I^{sqr}_{ij,t,d} + \sum_{s=1}^{S} P^c_{j,t,d,s} = \sum_{r=1}^{R} m^S_{ij,r} \Delta^P_{ij,t,d,r} + \sum_{r=1}^{R} m^S_{ij,t,r} \Delta^Q_{ij,t,d,r}$$
$$\forall ij \in \Omega_l, \forall t \in \Omega_t, \forall d \in \Omega_d \tag{30}$$

Constraints (24), (26), (28) and (30) replace (15). Note that the number of operation variables has been increased with the linearization, while the number of investment variables does not change. Further, as it will be illustrated in Section 3, this kind of optimization problem can be solved with the help of standard commercial solvers. Note also that (12)–(14), (24), (26) and (28) are linear expressions that represent the steady-state operation of a radial distribution system.

3. Tests and results

To solve the DGA problem through the proposed formulation, a modified 33-bus radial distribution test system of 12.66 kV with real and reactive power demand of 5325 kW and 3600 kVAr, respectively, was used. Original information with regard to this test system can be found in [27]. The model of the DGA problem has

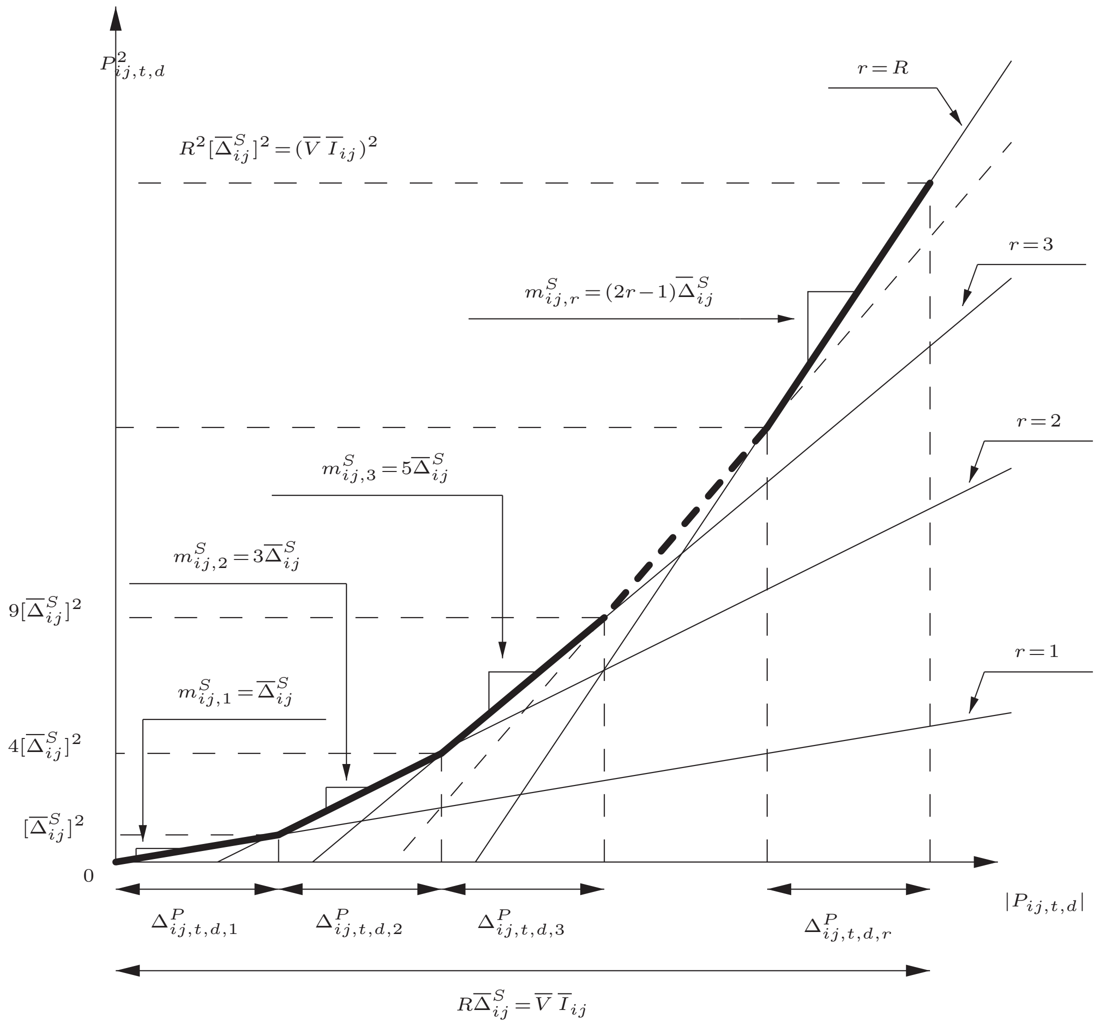

Fig. 5. Modeling the piecewise linear P^2_{ij} function.

been implemented in AMPL [28] and solved with CPLEX [29] using a workstation PC with an Intel *i*7 860 processor.

Three load levels (Ld_1, Ld_2, and Ld_3), for two system topologies (T_1 and T_2) were considered to perform seven types of tests, as shown below:

- Base case: without DGs
- S-C, DI-C and SDI-C: with DGs to be connected to the network through only SGs, only DFIGs, and SGs or DFIGs, respectively, but regardless of short-circuit constraints
- S-C_{cc}, DI-C_{cc} and SDI-C_{cc}: with DGs to be connected to the network through only SGs, only DFIGs, and SGs or DFIGs, respectively, considering short-circuit constraints

For each DG type, two different capacities were supposed. Thus, DGs connected to the network through SGs and with capacities of 1000 and 2000 kW are labeled as DG_1 and DG_2, and DGs connected to the network through DFIGs and with capacities of 1000 and 2000 kW are labeled as DG_3 and DG_4. Additionally, it was supposed that $\overline{n}_{dg} = 2$.

Substation energy cost of 0.15, 0.10 and 0.08 US$/kWh for the load levels Ld_1, Ld_2 and Ld_3 (with percentages around 100%, 50% and 30% of nominal demand), respectively, were assumed. Further, times of 1000, 6760 and 1000 h for the load levels Ld_1, Ld_2 and Ld_3, and percentages of 80% and 20% of the total time in each load level for topologies T_1 and T_2, respectively, were assumed.

Investment costs of 200×10^3, 380×10^3, 170×10^3 and 340×10^3 US$ and subtransient reactances of 0.032, 0.016, 0.048 and 0.024 Ω, for DG_1, DG_2, DG_3 and DG_4, respectively, were considered. The equivalent external reactance and the minimum leading and lagging power factor of the substation were assumed as 0.021 Ω and 0.8, respectively. Regarding to the linearization parameters, the following values were used: $S=4$, $R=20$ and $A=6$.

To analyze only the influence of the *coupling element* used to connect the DGs to the network (SG or DFIG), a constant DG energy cost of 0.02 US$/kWh for both DG technologies was assumed.

All parameters assumed correspond to typical values, which were chosen in such a way that comparative analysis can be performed.

In the base case, the total cost is 2383.9×10^3 US$, and the losses are 109.79, 26.91 and 9.75 kW for topology T_1, and 113.32, 28.01 and 10.16 kW for topology T_2, for the load levels Ld_1, Ld_2, and Ld_3, respectively. Simulation results related to types, sizes, allocation buses, total costs, run times and losses for the other cases are presented in Table 2. As shown in this table, all cases improve the base case in terms of costs and losses for all load levels and topologies. Average percentages of improvement of the base case regarding the losses are 68.18%, 67.34% and 68.84% for topology T_1, and 62.78%, 65.63% and 68.39% for topology T_2, for the load levels Ld_1, Ld_2, and Ld_3, respectively. Regarding the total costs, the average percentage of improvement is 46.24%. Cases SDI-C and SDI-C_{cc} are those with

Table 2
Simulation results.

	S-C	DI-C	SDI-C	S-C_{cc}	DI-C_{cc}	SDI-C_{cc}
Allocation bus (DG_{Type} selected)	5(DG_2) 23(DG_1)	2(DG_4) 28(DG_4)	4(DG_2) 29(DG_3)	5(DG_2) 23(DG_1)	7(DG_4) 23(DG_4)	12(DG_3) 22(DG_2)
Costs (10^3 US$)	1258.80	1335.98	1249.23	1258.80	1336.24	1250.37
Run time (min)	11	5	27	56	58	517
Losses (kW)						
Ld_1						
T_1	33.27	28.74	34.30	33.27	34.47	45.58
T_2	40.65	53.19	48.88	40.65	30.42	39.29
Ld_2						
T_1	7.67	6.38	9.15	7.67	10.26	11.61
T_2	9.66	12.43	10.36	9.66	7.83	7.83
Ld_3						
T_1	2.69	2.13	3.18	2.69	3.52	4.02
T_2	3.04	4.02	3.63	3.04	2.74	2.80

Table 3
Simulation results using the AC power flow.

	S-C	DI-C	SDI-C	S-C_{cc}	DI-C_{cc}	SDI-C_{cc}
Costs (10^3 US$)	1258.83	1336.00	1249.25	1258.83	1336.25	1250.39
Losses (kW)						
Ld_1						
T_1	33.37	28.88	34.40	33.37	34.53	45.70
T_2	40.83	53.34	48.98	40.83	30.55	39.48
Ld_2						
T_1	7.77	6.42	9.13	7.77	10.26	11.64
T_2	9.72	12.55	10.42	9.72	7.90	7.81
Ld_3						
T_1	2.67	2.06	3.08	2.67	3.49	4.03
T_2	3.01	3.97	3.60	3.01	2.69	2.71

Table 4
Differences with respect to the AC power flow (in percentages).

	S-C	DI-C	SDI-C	S-C_{cc}	DI-C_{cc}	SDI-C_{cc}
Costs	2.38×10^{-3}	1.50×10^{-3}	1.60×10^{-3}	2.38×10^{-3}	7.48×10^{-3}	1.60×10^{-3}
Losses						
Ld_1						
T_1	0.30	0.48	0.29	0.30	0.17	0.26
T_2	0.44	0.28	0.20	0.44	0.43	0.48
Ld_2						
T_1	1.29	0.62	0.22	1.29	0.00	0.26
T_2	0.62	0.96	0.58	0.62	0.89	0.26
Ld_3						
T_1	0.75	3.40	3.25	0.75	0.86	0.25
T_2	1.00	1.26	0.83	1.00	1.86	3.32

lower total costs when compared with their analogous cases with and without considering short-circuit constraints.

To show the accuracy of the proposed MILP model, power losses and costs of the obtained solutions (presented in Table 2) were compared with power losses and costs calculated using an AC power flow, which are presented in Table 3. These results show that the maximum differences (from Table 4) between the values obtained through the AC power flow and the result obtained through the linearized model are only 7.48×10^{-3}% for costs (case DI-C_{cc}) and 3.4% for losses (case DI-C for topology T_1 and load level Ld_3).

Table 5
Costs for cases S-C, DI-C, DI-C* and SDI-C.

	S-C	DI-C	DI-C*	SDI-C
Investment cost (10^3 US$)	580	680	510	550
Substation energy cost (10^3 US$)	285.30	259.06	523.70	308.60
DGs energy cost (10^3 US$)	393.50	396.92	350.60	390.63
Total costs (10^3 US$)	1258.80	1335.98	1384.30	1249.23

Attempts to solve the non-linear model of Section 2.5 for all cases with DGs, i.e., for cases S-C, DI-C, SDI-C, S-C_{cc}, DI-C_{cc} and SDI-C_{cc}, were performed using the MINLP solver KNITRO [30]. In neither case the solver KNITRO succeeded in finding the optimal solution. As an example, with a simulation time limit of 1000 min, a solution was found for case S-C (which is the simplest case) with costs of 1258.85×10^3 US$, which is a worse solution than the one found with the proposed method for the same case (1258.80×10^3 US$ in 11 min) and at a longer time in about 100 times. Thus, we can conclude that the proposed MILP method is efficient in comparison with the MINLP method.

Note that only in cases DI-C and DI-C_{cc} (from Table 2) the optimal solution of the problem considers two generators with maximum capacity (2000 kW) to be installed; in addition, these cases have the highest costs: 1336.0×10^3 and 1336.2×10^3 US$, respectively. It might be thought that, if, in the case DI-C, the result was similar to the other cases without short-circuit constraints (cases S-C and SDI-C), namely, if the optimal solution was to install generators of 1000 and 2000 kW, then the total costs would decrease. However,

Table 6
Impedances between DGs.

	DI-C		SDI-C		DI-C_{cc}		SDI-C_{cc}	
	T_1	T_2	T_1	T_2	T_1	T_2	T_1	T_2
Impedance (Ω)	1.9259	2.3254	1.7648	2.0442	2.4135	3.4014	2.4135	4.3343

factors such as reduced ability to deliver real and reactive power in the first operation quadrant of the DFIGs, when they are compared to the SGs, and the constraint (20), that limits the reactive power delivered by the substation to the system, do not allow that the optimal solution is one without considering the two generators with maximum capacity. Thus, in the case DI-C, the optimal solution indicates that it is better to increase the investment cost of installing two generators with maximum capacity, paying less to the substation for power generation, than any other solution of reducing investment costs by installing generators of 1000 and 2000 kW and paying more to the substation for power generation.

To illustrate the above observation, a test on an alternate case, similar to case DI-C, identified here as case DI-C*, in which only one generator of each capacity can be installed, was done. The results related to the costs of this case, in addition to the costs of cases S-C, DI-C and SDI-C, are presented in Table 5.

The optimal solution obtained for case DI-C* indicates a total cost of 1384.3×10^3 US$, which is higher than the cost of cases S-C, DI-C or SDI-C. Although the investment and DGs energy costs in case DI-C* are lower than those presented in cases S-C and SDI-C, the power generation cost of the substation is the value that makes the solution of case DI-C* more expensive than the other cases.

Capability curves of the DGs for the first operation quadrant (providing reactive power) represented by their linear equivalents and operation points of the optimal solutions of cases S-C, DI-C, and SDI-C, as well as case DI-C*, for Ld_1 and T_1, are depicted in Fig. 6. In this figure, note that the limited operation of the DFIGs with respect to the SGs, which is shown in the gray filled area, is evidenced.

With regard to the short-circuit constraints, the importance of their inclusion is reflected specifically in cases DI-C and SDI-C, where the short-circuit capacity is exceeded by 3.7% and 0.8% in T_1 and T_2 for case DI-C, and by 3.7% and 2.3% in T_1 and T_2 for case SDI-C, respectively. In cases DI-C_{cc} and SDI-C_{cc}, as a result of the inclusion of these constraints, the short-circuit currents were brought back to the allowed limits through the reallocation of the DGs. As presented in Table 6, for all cases in which the DGs were reallocated to consider the short-circuit constraints, the trend is to reallocate the DGs seeking to increase the impedance of the path of the system followed from one DG to another.

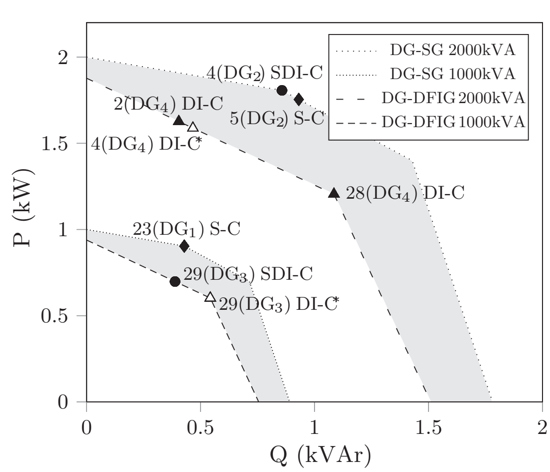

Fig. 6. Operation points of cases S-C, DI-C, DI-C* and SDI-C for Ld_1 and T_1.

4. Conclusions

A novel MILP approach to solving the problem of optimal type, size and allocation of DGs in radial distribution systems was presented. This approach is a useful tool for distribution systems planning to help take appropriate decisions in accordance with the characteristics of the system under study and economic resources available. The results demonstrate that, through the proposed linearized model, solutions can be obtained with high efficiency using a conventional MILP solver. Furthermore, the solutions delivered by the presented proposal are obtained in reasonable computation times.

In the proposed formulation, the steady-state operation of radial distribution systems were modeled. The presented mathematical model is much closer to reality compared to previous works on the matter of allocation of DGs, because we consider the capability curves of the *coupling element* of the DGs and short-circuit currents constraints in the mathematical model. At the same time, we transform the non-linear model into a linear model to be solved through a commercial solver, which use is easy and fast.

For the 33-bus radial distribution test system, two different topologies, three load levels and two types of DGs represented by the capability curves of SGs and DFIGs were considered. The results show the accuracy as well as the efficiency of the proposed solution technique. Furthermore, the results illustrate the importance of modeling the *coupling element* of the DGs in order to have their true capacity as well as the short-circuit constraints to ensure compliance with the short-circuit capacity of the system.

Another important contribution of this work is that we are introducing a new way to perform the linearization of the power flow constraints, which presents an efficient computational behavior with conventional MILP solvers. With appropriate considerations, the presented linearization method could even be used for other types of analysis, such as those performed in transmission systems, where there are also non-linear expressions that can influence the efficiency of the power flow.

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